Semi-Supervised Semantic Remote Sensing Image Change Detection Using Multimodal Spatiotemporal Association Knowledge

Linye Zhu, Huaqiao Xing, Wenbin Sun, Shouhang Du, and Deqin Fan

***Abstract*—Affected by the cost of producing semantic change detection samples, semi-supervised semantic change detection methods have gradually become a research hotspot. However, existing semi-supervised semantic change detection methods only consider unimodal remote sensing image data, which easily overlooks the rich spatiotemporal semantic information available in the data. This limitation restricts the performance of semi-supervised semantic change detection. To address these issues, this study proposes a multimodal spatiotemporal association knowledge-based semi-supervised semantic change detection (MST-AK) method. First, the MST-AK method introduces the CLIP model for the first time to construct land cover descriptive textual knowledge, overcoming the limitations of unimodal data in semi-supervised semantic change detection. Second, the image–text dual-attention difference change feature fusion module and the three-branch interactive spatiotemporal association module are designed to enable cross-modal learning between different spatiotemporal images from a multimodal perspective. Finally, the change magnitude-based adaptive threshold selection method is used to adaptively select the threshold value based on the distribution of change magnitude in the change feature image, improving the distinguishability of the semantic change detection results. To demonstrate the feasibility of the MST-AK method, two commonly used publicly available semantic change detection datasets, SECOND and Landsat-SCD, are utilized for comparison and validation against state-of-the-art methods. In addition, areas in Beijing City and Yangzhou City, China, are selected to demonstrate the application of semantic change detection. The results show that the MST-AK method can fully explore the temporal and spatial semantic change features of multimodal remote sensing images, break the limitations of traditional models relying on a single modality, enable cross-modal learning and understanding, and improve the accuracy of semantic change detection results even with fewer samples.**

***Index Terms*—Domain knowledge, remote sensing images, semantic change detection, semi-supervised method, spatiotemporal association.**

I. Introduction

REMOTE sensing image semantic change detection refers to the process of using remote sensing images to identify detailed change type information in addition to obtaining change areas [1], [2]. Compared to binary change detection, semantic change detection is more challenging but provides more comprehensive information, including the conversion of land cover types (i.e., "from-to" transition information) in the study area [3], [4]. This makes it more suitable for application areas such as urban expansion monitoring, post-disaster assessment, and land use analysis [5], [6].

Compared to binary change detection labeled samples, semantic change detection labeled samples are more expensive to produce. In practice, the area of changed regions is much smaller than that of unchanged regions [7], [8]. Semantic change detection datasets typically characterize land cover types only for the changed regions, rather than for the entire image, which makes the task more challenging [9]. Traditional semantic change detection methods can be divided into direct comparison and post-classification comparison methods [10], [11]. The direct comparison method treats each type of change as a separate category. This often leads to a proliferation of output categories, which negatively impacts change type recognition accuracy. The post-classification comparison method involves comparing the post-classification results of bitemporal images. However, this method is prone to error accumulation due to inaccuracies in the classification results [12]. With the development of deep learning, deep learning-based semantic change detection methods have become a hot research topic due to their powerful feature extraction capabilities, which outperform traditional methods. Existing semantic change detection methods can be categorized as follows [13].

Single-Branch-Based Semantic Change Detection Methods: This type of method primarily involves a single-output branching Siamese network, which directly inputs the difference map or connection map of the bitemporal image into the network. It then extracts feature information through the weight-sharing encoder and inputs it into the semantic change detection network decoder. For example, Daudt et al. [14] proposed the fully convolutional Siamese-concatenation (FC-Siam-conc) and fully convolutional Siamese-difference (FC-Siam-diff) methods for change detection in multitemporal images. Mou et al. [15] proposed a recursive convolutional neural network (ReCNN) architecture, trained to learn joint spectral–spatial–temporal feature representations within a unified framework for change detection in remote sensing images. This type of method treats each potential change type as a separate category and approaches semantic change detection as a semantic segmentation task. When considering the number of land cover types, the number of change types grows quadratically, leading to class imbalance issues that negatively impact semantic change detection

Received 1 June 2025; revised 13 July 2025 and 12 September 2025; accepted 13 October 2025. Date of publication 15 October 2025; date of current version 4 November 2025. This work was supported by the National Key Research and Development Program of China under Grant 2023YFB3907600. *(Corresponding author: Huaqiao Xing.)*

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Digital Object Identifier 10.1109/TGRS.2025.3621732

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results. Furthermore, this method does not fully leverage cross-modal prior knowledge. These approaches overlook the complementary role of textual modalities in remote sensing semantics, such as descriptive knowledge of land cover types, which can significantly enhance the interpretability of change understanding.

Dual-Branch-Based Semantic Change Detection Methods: This type of method performs separate classification of bitemporal images and obtains change type information by comparing the two classification results. For example, Peng et al. [16] proposed a new large-scale semantic change detection convolutional network (SCDNet) that introduces an attention mechanism and a deep supervision strategy to improve network performance. Xia et al. [17] proposed a deep Siamese post-classification fusion network (PCFN) that efficiently integrates the temporal correlation between two images by processing joint features to determine changes and identify specific change types through automatic soft fusion. The accuracy of semantic change detection results for this method is largely determined by the accuracy of the classification results and is prone to error accumulation, especially in boundary regions, which in turn affects the detection results. Furthermore, such methods suffer from insufficient feature representation capabilities due to interference from external factors. Remote sensing image change detection has strong spatiotemporal correlation, which is easily affected by external factors and can lead to misjudgment of land cover types. The traditional unimodal method relies solely on image data, which may make it difficult to fully utilize the spatiotemporal semantic association information, leading to change misjudgments.

Multitask-Based Semantic Change Detection Methods: This type of method learns multiple tasks simultaneously, including bitemporal image feature extraction or semantic classification and bitemporal image binary change detection, which ultimately produces semantic change detection results. For example, Daudt et al. [18] proposed a framework that performs both change detection and land cover mapping, using predicted land cover information to help predict changes. Ding et al. [19] proposed a semantic change detection method that merges semantic temporal features into a deep change detection unit for reasoning about bitemporal semantic correlations. Zheng et al. [20] proposed a deep multitask encoder–transformer–decoder architecture (ChangeMask) to obtain semantic segmentation and binary change detection results by exploring semantic change causality and temporal symmetry. Chen et al. [21] proposed a dual encoder–decoder network backbone to construct a dense connection-based feature fusion module and a self-supervised learning-based strategy to obtain semantic segmentation results without using semantic segmentation labels. Niu et al. [22] proposed a new symmetric multitasking network (SMNet) integrating global and local information for semantic change detection. Long et al. [23] proposed a hierarchical semantic graph interaction network (HGINet) for semantic change detection of high-resolution remote sensing images, combining semantic segmentation and change detection tasks. Tan and Zuo [24] proposed a change-aware-guided multitasking network (CGMNet) that integrates change-aware mask branching and utilizes prior knowledge of change regions to enhance land cover classification in bitemporal remote sensing images. Mei et al. [25] proposed the SCD-segment anything model (SAM) method, which aims to utilize the powerful visual recognition function of SAM to improve the accuracy of semantic change detection. Compared with the previous two types of methods, the semantic change detection results of this type of methods are more accurate and stable.

Affected by the number of semantic labeled samples, the semi-supervised learning-based semantic change detection method has become a research hotspot [26]. This method obtains semantic change detection results through a small number of semantic labeled samples and a large amount of unlabeled image data. For example, Wang et al. [27] proposed an enhanced multitask semantic change detection method with semi-supervised learning for land cover segmentation, which effectively utilizes a large amount of unlabeled land cover data and improves the overall performance of the multitask framework. Heidler et al. [28] proposed a semi-supervised learning approach for PixelDINO to improve model generalization across the Arctic with a limited number of labels. Lv et al. [29] proposed a semi-supervised semantic and spatially varying detail retention network (S3CDRNet) for accurately modeling spatially varying details and distributions of multiple change classes in the remote sensing images. Zhang et al. [30] proposed a semi-supervised contrast learning (SSCLNet) method and designed a semi-supervised contrast learning module for semantic segmentation to enhance the distinguishability of categories. Chen et al. [31] proposed a cross-consistency-guided semi-supervised learning semantic segmentation network for high-quality multicategory semantic segmentation of complex remote sensing scenes with a limited number of labeled images. Dai et al. [32] proposed a difference enhancement and class-aware rebalancing semi-supervised network (Semi-DECRNet) for farmland semantic change detection. The above studies improve the accuracy of semi-supervised semantic change detection results in different ways. However, such methods lack adaptability in threshold selection, making it difficult to effectively detect changes. These methods usually rely on a single threshold, and the lack of adaptive threshold selection based on the magnitude of changes in the image limits further improvement of semi-supervised semantic change detection accuracy.

Combined with the above analysis, the main purpose of semi-supervised semantic change detection methods is to utilize a small number of labeled samples and a large number of unlabeled images to deeply mine the available spatiotemporal semantic features, improve the semantic correlation between different temporal branches, and co-construct a semantic change detection model [30].

With the rapid development of large-scale visual-linguistic models, they have shown unique advantages in the field of semantic change detection in remote sensing. Visual-linguistic representation learning has revolutionized change detection by establishing a deep association between visual features and semantic concepts through comparative learning on large-scale image–text pairs [33]. Traditional methods are often limited to pixel-level difference analysis, making it difficult to distinguish real semantic changes from representational changes. Visual-linguistic representation learning models, especially the CLIP

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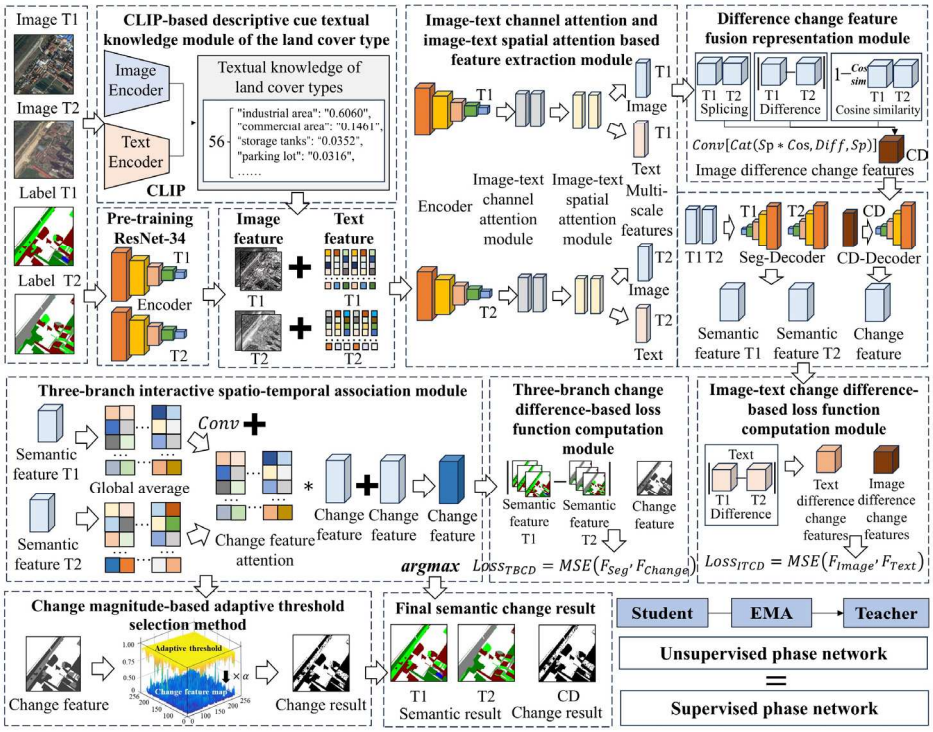


Fig. 1. Framework of the MST-AK method. The MST-AK method mainly consists of the pretraining weights of the ResNet-34 network, the CLIP-based descriptive cue textual knowledge module of the land cover type, the image–text channel attention-based and image–text spatial attention-based feature extraction module, the difference change feature fusion representation module, the three-branch interactive spatiotemporal association module, the image–text change difference-based loss function computation module, the three-branch change difference-based loss function computation module, and the change magnitude-based adaptive threshold selection method.

model, can accurately capture land cover changes with practical significance through the natural language comprehension ability, which significantly reduces the false detections caused by seasonal changes, light differences, and other disturbing factors [34]. The open vocabulary feature of the CLIP model makes it perform exceptionally well in scenarios with few or even zero samples, and it can recognize new categories that have not appeared in the training data with only simple textual hints [35]. Meanwhile, its cross-modal alignment capability effectively alleviates the problem of distributional differences among multisource remote sensing data, allowing the model to maintain stable detection performance across data from different sensors and phases [36]. Moreover, the CLIP model not only relies on the changes between image pixels but also understands the semantics of these changes, which enhances the interpretability of the results and realizes the multimodal semantic correlation between image and text [37].

Therefore, this article proposes a multimodal spatiotemporal association knowledge-based semi-supervised semantic change detection (MST-AK) method. The core idea of the MST-AK method is to fully exploit the descriptive cue text knowledge of remote sensing images based on 56 common land cover types using the CLIP model from a multimodal perspective. It combines the spectral–spatial difference change feature fusion representations of the images and texts to realize the three-branch interaction spatiotemporal semantic change feature extraction and jointly constructs a semi-supervised semantic change detection model. The main innovations and contributions of this study are as follows.

1) *Fusion of Multimodal Cross-Domain Knowledge:* The MST-AK method innovatively applies the CLIP model to the semi-supervised semantic change detection task in remote sensing images. It aims to integrate remote sensing images with textual descriptions to provide richer semantic context for land cover type changes. Overcoming the limitations of traditional single modality reliance, it enables change detection in images to be combined with broader descriptive information, thereby enhancing the depth and accuracy of semantic understanding.
2) *Multilevel Exploration of Spatiotemporal Semantic Change Features:* By integrating image and textual information, the MST-AK method conducts an in-depth exploration of semantic change features in remote sensing images from spatiotemporal dimensions, particularly focusing on interactions between different temporal images. Through cross-modal learning, it better captures and interprets semantic changes in remote sensing images, further enhancing spatiotemporal semantic correlations and enabling more precise detection of land cover changes.
3) *Adaptive Threshold Optimization Based on Change Magnitude:* The MST-AK method proposes an adaptive threshold selection strategy based on change magnitude, automatically adjusting thresholds according to the magnitude distribution of change feature images. This effectively optimizes the threshold selection process, enhancing the model's robustness across diverse

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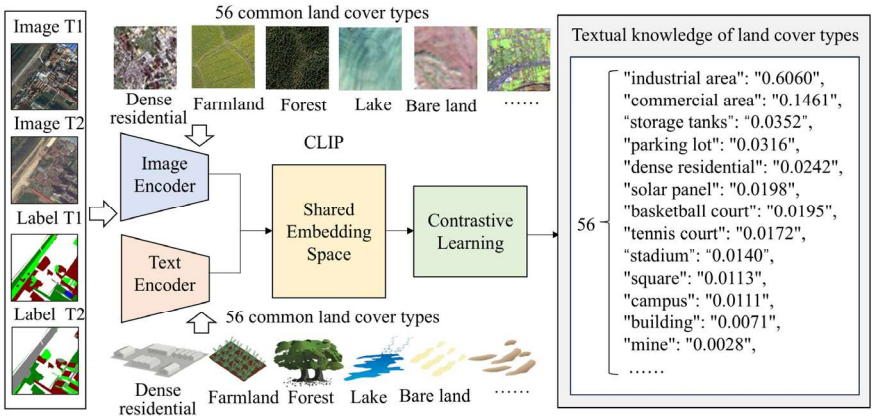


Fig. 2. CLIP-based descriptive cue textual knowledge module of the land cover type. Using the CLIP model to perform unsupervised text semantic feature analysis on semantic change detection data across different time phases, thereby obtaining text semantic prediction results for images.

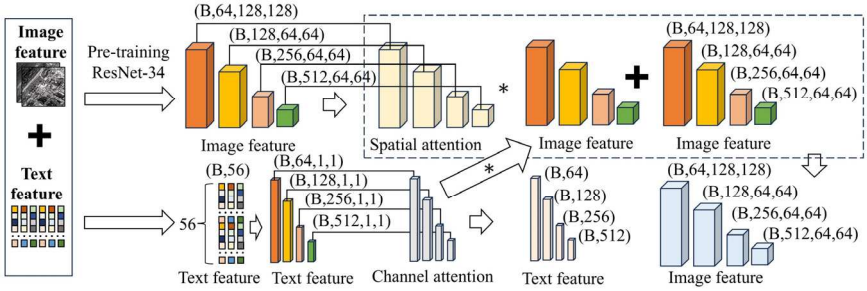


Fig. 3. Image–text channel attention-based and image–text spatial attention-based feature extraction module. By dynamically integrating visual and textual features through an image–text cross-modal attention mechanism, it highlights key channels and spatial regions relevant to change detection.

change scenarios and improving the accuracy of semantic change detection.

4) *Practical Application Validation and Scenario Adaptability:* To validate the practical value of the MST-AK method, experiments conducted on remote sensing images from selected areas in Beijing City and Yangzhou City further demonstrated its effectiveness in real-world scenarios. In addition, experiments using publicly available semantic change detection datasets confirmed the method's adaptability and feasibility across diverse data environments (source code available at https://github.com/zhuzhu94854693/MST-AK).

The rest of this article is organized as follows. Section II describes the specifics of the MST-AK method. Section III analyzes the experimental results of the MST-AK method. Section IV is the discussion and application demonstration. Section V is the conclusion and future research.

II. Methodology

This section introduces the specific framework details of the MST-AK method. As shown in Fig. 1, the MST-AK method mainly consists of the pretraining weights of the ResNet-34 network, the CLIP-based descriptive cue textual knowledge module of the land cover type, the image–text channel attention-based and image–text spatial attention-based feature extraction module, the difference change feature fusion representation module, the three-branch interactive spatiotemporal association module, the image–text change difference-based loss function computation module, the three-branch change difference-based loss function computation module, and the change magnitude-based adaptive threshold selection method.

A. Overview

To fully utilize a small number of labeled samples and a large number of unlabeled images, the MST-AK method consists of two phases. The first phase is a supervised phase, where only labeled samples are used for training to obtain usable pretrained weight models. The second stage is the unsupervised stage, where both labeled samples and unlabeled images are used for training. In this phase, the student model extracts the semantic change feature information, the teacher model generates the corresponding pseudo-semantic labels, and the parameters of the teacher model are updated using the exponential moving average of the student model.

B. MST-AK Semantic Change Network

The MST-AK semantic change network deeply mines spatiotemporal semantic change features from an image–text multimodal perspective to realize three-branch interactive spatiotemporal semantic associations between different temporal images.

1) *The Clip-Based Descriptive Cue Textual Knowledge Module of the Land Cover Type:* To enhance the discriminative power for complex scenes, both pixel-level visual differences and high-level semantic changes are considered. The graphic and textual cross-modal comprehension ability of the CLIP model is utilized to

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provide a priori textual knowledge guidance for semi-supervised semantic change detection in remote sensing images. As shown in Fig. 2, for the first time, the CLIP image–text macromodel is combined with a semi-supervised semantic change detection method for remote sensing images to provide descriptive cue textual knowledge for a pixel-level semantic change detection dataset. Specifically, by combining 56 common land cover types, the CLIP model [34] is used to unsupervisedly judge the textual semantic features of different simultaneous phase semantic change detection data, obtain the textual semantic prediction results of the images, and prepare for the subsequent extraction of different simultaneous multimodal spatiotemporal semantic change features.

2) *The Image–Text Channel Attention-Based and Image–Text Spatial Attention-Based Feature Extraction Module:* To achieve semantically oriented screening of feature channels, locate spatially semantically sensitive regions, and hierarchically fuse multilevel features to address the complexity of the remote sensing scenes, we dynamically fuse visual and textual features through the mechanism of image–text cross-modal attention. This approach highlights the key channels and spatial regions related to change detection. As shown in Fig. 3, this module mainly consists of the image–text channel attention module, the image–text spatial attention module, and the multiscale multimodal feature extraction module. The image–text channel attention module is designed to guide the adjustment of channel weights based on the input text features (e.g., category confidence), and learns the weights of each channel through a series of linear transformations of the input text features, thus strengthening or suppressing the features of a particular channel. The image–text spatial attention module computes spatially important regions based on the input feature map and outputs a spatial attention map representing the importance of each location in the feature map by dimensionalizing the input features using a simple 1×1 convolution kernel. The multiscale multimodal feature extraction module contains several image–text channel attention modules and image–text spatial attention modules, which act on feature maps at different scales. The input image features and text features are weighted by each attention module and combined with a residual connection to output the final features.
3) *The Difference Change Feature Fusion Representation Module:* To accurately distinguish real surface changes from disturbance noise, robust change representations are constructed by calculating and fusing multiangle and multiscale differential features.

This module is mainly used to calculate multiscale difference change features and perform multiscale change feature fusion. From the perspectives of feature value and feature shape, the feature splicing method, absolute difference method, and cosine similarity method are utilized to jointly obtain the multiscale difference change features.

The feature splicing method is formulated as

$$\mathrm{FS} = \mathrm{Feature}_1 - \mathrm{Feature}_2. \tag{1}$$

The absolute difference method is formulated as

$$\mathrm{AD} = |\mathrm{Feature}_1 - \mathrm{Feature}_2|. \tag{2}$$

The cosine similarity method is formulated as

$$\mathrm{CS} = 1 - \cos_\mathrm{sim}\left(\mathrm{Feature}_1, \mathrm{Feature}_2\right). \tag{3}$$

On this basis, the multiscale difference change features acquired from different perspectives are spliced, and convolved feature fusion is performed to obtain the final multiscale change features. The final multiscale change features are formulated as

$$\mathrm{Final}_{\mathrm{CF}} = \mathrm{Conv}\left[\mathrm{Cat}\left(\mathrm{FS} * \mathrm{CS}, \mathrm{AD}, \mathrm{FS}\right)\right] \tag{4}$$

where $\mathrm{Feature}_1$ and $\mathrm{Feature}_2$ are the multiscale feature extraction maps of the two temporal phases. FS is the difference change feature map of feature splicing method. AD is the difference change feature map of absolute difference method. CS is the difference change feature map of the cosine similarity method. $\mathrm{Final}_{\mathrm{CF}}$ is the final multiscale change feature map. Conv denotes the convolution operation.

4) *The Three-Branch Interactive Spatiotemporal Association Module:* To enhance the temporal correlation and spatial contextual understanding of land cover semantic change, multimodal spatiotemporal interaction modeling is utilized.
5) *Loss Function:* In the supervised and unsupervised phases, the semantic segmentation loss function, binary change detection loss function, semantic consistency loss function, image–text change difference-based loss function computation module, and three-branch change difference-based loss function computation module are used to compute the loss between the predicted image and the true label or pseudo-label.
6) *The Change Magnitude-Based Adaptive Threshold Selection Method:* By dynamically adjusting the pixel-level change detection thresholds, it solves the shortcomings of traditional fixed threshold methods in complex scenes, thus significantly improving the accuracy and adaptability of change detection. As shown in Fig. 4, this method adaptively selects a corresponding threshold for each pixel of the change feature image, selecting a smaller threshold for regions that may have changed and a larger threshold for regions that may not have changed, to improve the adaptability of change threshold selection and avoid omissions and misrecognition in semantic change detection. Details can be found in [38]. The specific formulas are as follows:

This module flexibly adjusts the weights of different parts through the introduction of global semantic information, an attention mechanism, and a residual connection to strengthen the network's focus on spatiotemporal semantic changes, thereby improving the model's understanding and performance of temporal changes. The spatial mean of the input data is computed to capture spatiotemporal semantic variations for subsequent processing. The specific formula is as follows:

$$\mathrm{mean} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathrm{input}\left[:, :, i, j\right]. \tag{5}$$

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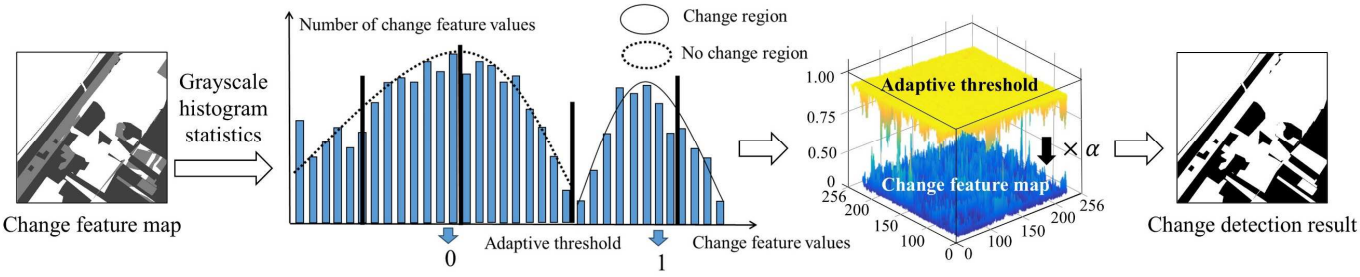


Fig. 4. Ideas of the change magnitude-based adaptive threshold selection method. By utilizing the change feature images, an adaptive threshold surface is constructed to select corresponding thresholds for each pixel. Smaller thresholds are assigned to regions that may have changed, while larger thresholds are assigned to regions that may remain unchanged.

By integrating mean information across different time points, the model dynamically adjusts input weights. The specific formula is as follows:

$$\text{weight} = \sigma\left(W_c \cdot (\text{mean1} + \text{mean2})\right). \tag{6}$$

The output combines a weighted version of input differences with the sum of raw input differences, enabling flexible spatiotemporal adjustments. The specific formula is as follows:

$$\text{Out} = \gamma \cdot \text{weight} \cdot \text{input}_{\text{diff}} + \text{input}_{\text{diff}} \tag{7}$$

where input is the feature extraction map for the two temporal phases. $\text{input}_{\text{diff}}$ is the difference change feature map. W_c is the weight matrix of the 1×1 convolution. σ is a sigmoid function. γ is a learnable scalar parameter.

5) *Loss Function:* In the supervised and unsupervised phases, the semantic segmentation loss function Loss_{SS}, binary change detection loss function Loss_{BCD}, semantic consistency loss function Loss_{SC}, image–text change difference-based loss function computation module $\text{Loss}_{\text{ITCD}}$, and three-branch change difference-based loss function computation module $\text{Loss}_{\text{TBCD}}$ are used to compute the loss between the predicted image and the true label or pseudo-label.

The semantic segmentation loss function Loss_{SS} is mainly a multiclass cross-entropy loss function computed on the prediction results of bitemporal semantic segmentation branches. The specific formula is as follows:

$$\text{Loss}_{\text{SS}} = -\frac{1}{N}\sum_{i=1}^{N} Y_{\text{True}_i/Pse_i} \log\left(Y_{\text{Pre}_i}\right) \tag{8}$$

where N is the number of land cover semantic types, in which $N = 0$ is the unchanged area; Y_{True_i/Pse_i} is the true labeled image or pseudo-labeled image; and Y_{Pre_i} is the predicted result image. The binary change detection loss function Loss_{BCD} directly optimizes the binary classification task for changed regions by supervising the signals, forcing the model to accurately distinguish between changed and unchanged pixels. The specific formulas are as follows:

$$\begin{aligned}\text{Loss}_{\text{BCD}} = &-\left(Y_{\text{True}/Pse}\log\left(Y_{\text{Pre}}\right)\right.\\ &+\left(1-Y_{\text{True}/Pse}\right)\log\left(1-Y_{\text{Pre}}\right)\Big)\\ &+\left(1-\frac{2Y_{\text{True}/Pse}\cdot\text{sigmoid}\left(Y_{\text{Pre}}\right)}{\sum}Y_{\text{True}/Pse}\right.\\ &\left.+\sum\text{sigmoid}\left(\text{Y}_{\text{Pre}}\right)\right)\end{aligned} \tag{9}$$

where Y_{True_i/Pse_i} is the true labeled image or pseudo-labeled image and Y_{Pre_i} is the predicted result image.

The semantic consistency loss function Loss_{SC} is mainly utilized to assess the probability distribution similarity of semantic change detection using the cosine function [19]. The specific formulas are as follows:

$$\text{Loss}_{\text{SC}} = \begin{cases}\cos\left(x_1, x_2\right), Y_{\text{True}/Pse} = 1\\ 1-\cos\left(x_1, x_2\right), Y_{\text{True}/Pse} = 0\end{cases} \tag{10}$$

where x_1 and x_2 are the feature vectors of the pixels at the same location on the bitemporal semantic segmentation branch.

The image–text change difference-based loss function computation module $\text{Loss}_{\text{ITCD}}$ mainly calculates the loss using image-based difference change features and text-based difference change features. It achieves cross-modality aligned change detection by jointly optimizing the difference features of visual and textual modalities to improve the model's semantic understanding of complex scenes and its anti-interference capability. The specific formulas are as follows:

$$\text{Loss}_{\text{ITCD}} = \text{MSE}\left(\text{F}_{\text{Image}}, F_{\text{Text}}\right) \tag{11}$$

where F_{Image} and F_{Text} are multiscale image difference change features and multiscale text difference change features extracted using MST-AK semantic change network, respectively.

The three-branch change difference-based loss function computation module $\text{Loss}_{\text{TBCD}}$ utilizes the bitemporal semantic segmentation branch for difference change extraction, which is combined with the binary change detection branch for loss calculation. It constructs a more robust representation of change features by co-optimizing the dual-supervised signals of semantic segmentation and binary change detection, thus addressing the limitations of using a single supervised signal. The specific formulas are as follows:

$$\text{Loss}_{\text{TBCD}} = \text{MSE}\left(\text{F}_{\text{Seg}}, F_{\text{Change}}\right) \tag{12}$$

where F_{Seg} and F_{Change} are the semantic segmentation difference change detection features and binary change detection features extracted using MST-AK semantic change network, respectively.

The total loss Loss consists of the above categories of losses. The specific formulas are as follows:

$$\text{Loss} = \frac{\left(\text{Loss}_{\text{SS}_1} + \text{Loss}_{\text{SS}_2}\right)}{2} + \text{Loss}_{\text{BCD}} + \text{Loss}_{\text{SC}} + \text{Loss}_{\text{ITCD}}$$

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TABLE I
RESULTS OF ACCURACY EVALUATION FOR DIFFERENT METHODS ON SECOND DATASET

Ratios	Method	Bi-SRNet	BT-HRSCD	CDSC	HGINet	SCanNet	SenseEarth2020	MST-AK
10%	mIoU	65.36	60.96	67.22	59.75	67.74	58.05	68.10
	Sek	10.38	4.49	12.98	4.66	-8.24	1.41	14.40
	Score	26.87	21.43	29.25	21.19	13.35	18.40	30.51
	Fscd	49.53	42.85	52.86	41.88	21.89	37.21	54.28
	Pre	50.65	43.48	56.93	51.07	21.49	34.72	49.90
	Rec	48.45	42.24	49.33	35.49	22.31	40.07	59.51
	OA	83.12	80.50	84.88	82.39	76.48	79.53	85.55
20%	mIoU	67.45	66.52	68.61	63.76	66.94	64.80	69.53
	Sek	13.88	12.26	15.31	9.10	12.83	9.08	16.67
	Score	29.95	28.54	31.30	25.50	29.07	25.80	32.53
	Fscd	53.90	51.62	55.39	48.46	52.62	47.45	56.67
	Pre	55.36	50.57	59.38	54.43	53.69	46.39	54.93
	Rec	52.52	52.70	51.91	43.67	51.58	48.57	58.53
	OA	84.57	83.23	85.64	83.68	84.10	82.58	85.69
30%	mIoU	68.97	68.21	69.67	65.38	68.22	66.30	70.54
	Sek	16.49	14.22	17.17	11.58	15.54	12.45	18.44
	Score	32.24	30.42	32.92	27.72	31.35	28.61	34.07
	Fscd	56.74	53.93	57.39	51.66	55.71	51.28	58.56
	Pre	57.64	54.56	61.06	54.48	54.58	54.31	56.67
	Rec	55.87	53.33	54.13	49.12	56.88	48.58	60.59
	OA	85.37	84.57	86.18	83.94	84.50	82.57	86.32
40%	mIoU	70.26	68.83	69.46	66.51	69.19	67.72	70.97
	Sek	17.96	15.29	17.25	13.77	16.92	13.40	18.68
	Score	33.65	31.35	32.92	29.59	32.60	29.69	34.36
	Fscd	57.84	54.79	57.67	54.44	57.04	52.00	58.56
	Pre	59.44	54.41	62.45	60.27	57.07	48.38	55.85
	Rec	56.33	55.18	53.58	49.63	57.01	56.21	61.54
	OA	86.03	84.65	86.37	85.29	85.29	84.88	86.59

$$+ \text{Loss}_{\text{TBCD}} + \alpha \cdot \text{Loss}_{\text{semi}} \tag{13}$$

where $\text{Loss}_{\text{semi}}$ is the loss function calculation of the pseudo-labeled image and the model prediction result. The calculation formula is consistent with the supervised phase formula, which is also $(\text{Loss}_{\text{SS}_1} + \text{Loss}_{\text{SS}_2})/2 + \text{Loss}_{\text{BCD}} + \text{Loss}_{\text{SC}} + \text{Loss}_{\text{ITCD}} + \text{Loss}_{\text{TBCD}}$. α is the parameter.

6) *The Change Magnitude-Based Adaptive Threshold Selection Method:* By dynamically adjusting the pixel-level change detection thresholds, it solves the shortcomings of traditional fixed threshold methods in complex scenes, thus significantly improving the accuracy and adaptability of change detection. As shown in Fig. 4, this method adaptively selects a corresponding threshold for each pixel of the change feature image, selecting a smaller threshold for regions that may have changed and a larger threshold for regions that may not have changed, to improve the adaptability of change threshold selection and avoid omissions and misrecognition in semantic change detection. Details can be found in [38]. The specific formulas are as follows:

$$\text{SCDM} = \begin{cases} \text{change}, & \text{CFM} \geq 1 - Y \cdot \alpha \\ \text{no_change}, & \text{CFM} < 1 - Y \cdot \alpha \end{cases} \tag{14}$$

where both CFM and Y are the change feature maps obtained using the MST-AK method. α is the parameter.

III. EXPERIMENTAL RESULTS AND ANALYSIS

In order to evaluate the effectiveness of the MST-AK method, publicly available datasets from the SECOND dataset and the Landsat-SCD dataset were used for comparative analysis with state-of-the-art semantic change detection methods. Six commonly used semantic change detection evaluation metrics [i.e., mean intersection over union (mIoU), separate Kappa (SeK), $F1$ score (Score), $F1$ score in segmentation and change detection task (Fscd), precision (Pre), recall (Rec), and overall accuracy (OA)] were utilized to evaluate the results of semantic change detection in terms of the corresponding precision [39], [40], [41]. The specific comparison methods are as follows.

1) *Bi-SRNet [19]:* This method merges semantic temporal features into deep change detection and contains two types of semantic inference blocks for reasoning about single-time and intertemporal semantic correlations to obtain semantic change detectionresults.
2) *BT-HRSCD [42]:* This method possesses two simple and effective modules. The first is a bidirectional shallow and deep feature aggregation module, and the second is a high-resolution difference extraction module to obtain semantic change detection results.
3) *CDSC [43]:* This method proposes a semantic change detection framework for cross-differential semantic coherence networks, which utilizes a 3-D cross-differential module combined with 3-D convolution, contrast learning, and deep semantic representation to obtain semantic change detection results.
4) *HGINet [23]:* This method proposes HGINet, designs a multilevel feature extractor, and quantifies the interaction between extracted features and different features of

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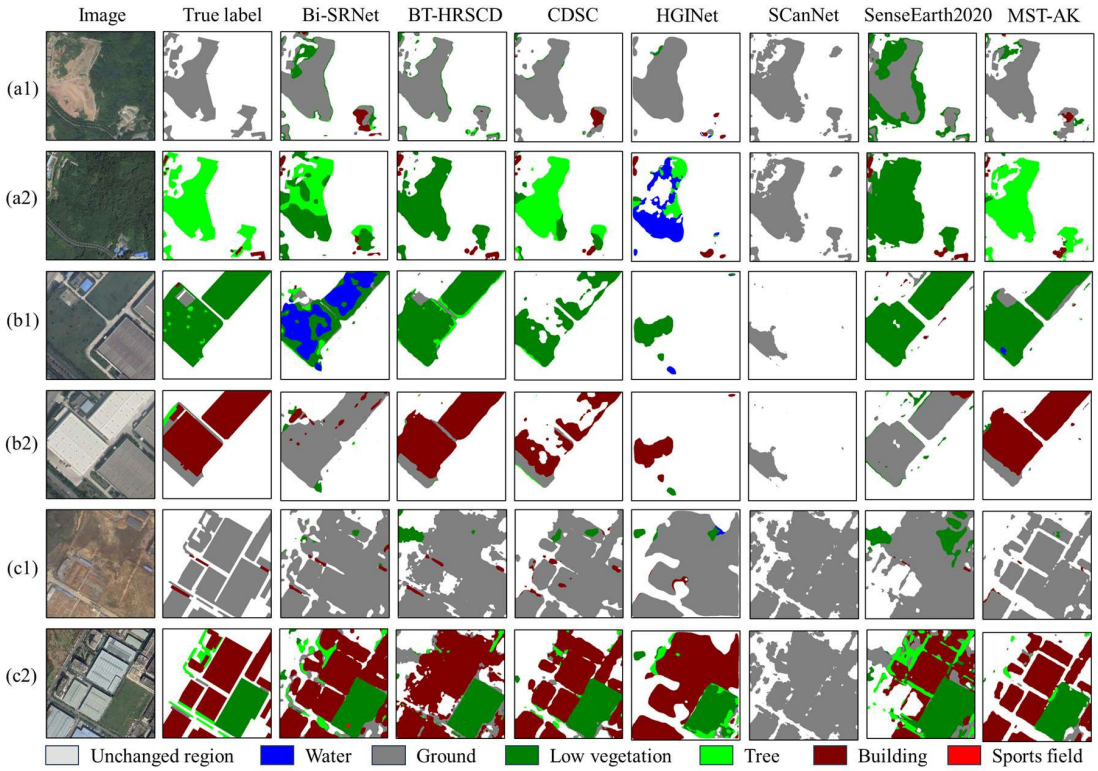


Fig. 5. Results of different methods for detecting changes in 10% of the samples in the SECOND dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

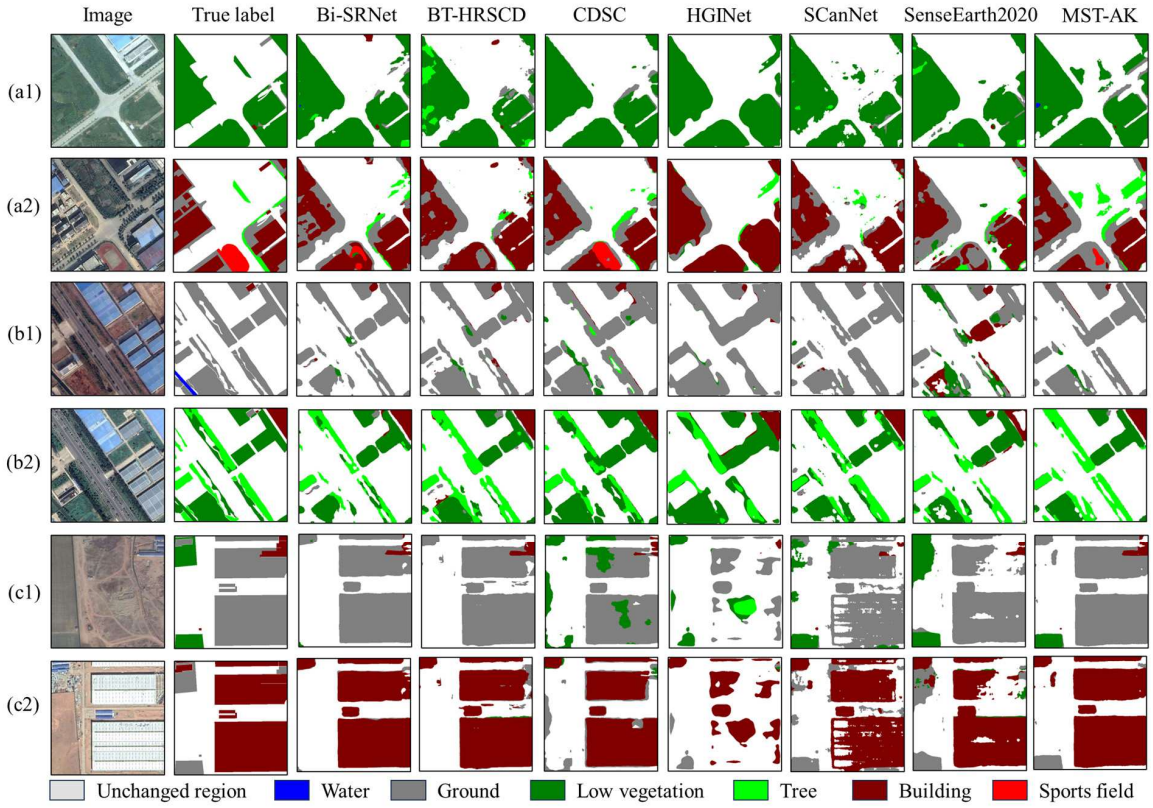


Fig. 6. Results of different methods for detecting changes in 20% of the samples in the SECOND dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

extracted difference features through graph learning to obtain semantic change detection results.

5) *SCanNet [44]:* This method proposes a semantic change converter and introduces a semantic learning

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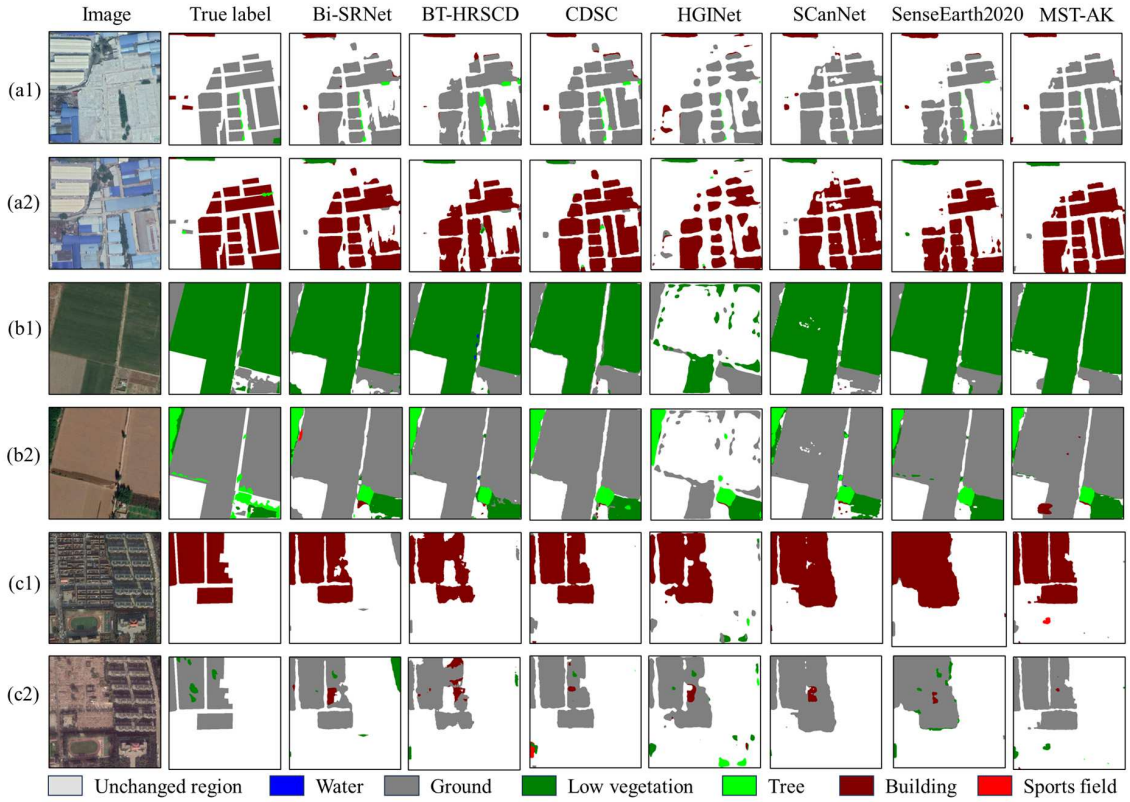


Fig. 7. Results of different methods for detecting changes in 30% of the samples in the SECOND dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

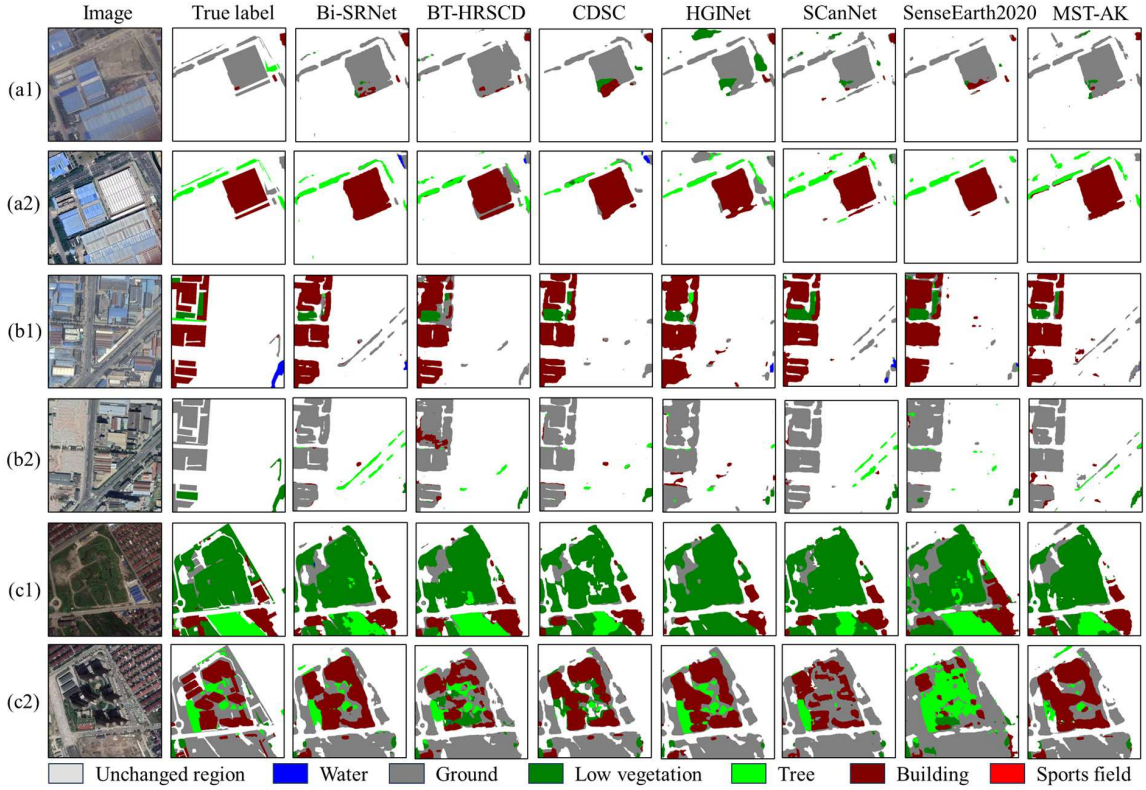


Fig. 8. Results of different methods for detecting changes in 40% of the samples in the SECOND dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

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TABLE II
RESULTS OF ACCURACY EVALUATION FOR DIFFERENT METHODS ON LANDSAT-SCD DATASET

Ratios	Method	Bi-SRNet	BT-HRSCD	CDSC	HGINet	SCanNet	SenseEarth2020	MST-AK
10%	mIoU	71.43	74.08	75.14	70.91	71.19	72.06	76.87
	Sek	21.31	25.08	27.27	21.76	20.53	20.28	30.78
	Score	36.34	39.78	41.63	36.50	35.73	35.81	44.61
	Fscd	63.82	66.61	68.98	64.17	63.15	61.99	71.69
	Pre	60.53	63.15	70.15	58.77	58.93	66.39	70.17
	Rec	67.50	70.48	67.85	70.67	68.02	58.13	73.29
	OA	83.56	85.04	86.79	82.96	83.06	83.08	87.96
20%	mIoU	75.48	77.49	78.93	76.85	79.39	75.61	82.28
	Sek	28.63	31.77	35.16	30.33	36.09	27.30	41.85
	Score	42.69	45.48	48.29	44.29	49.08	41.80	53.98
	Fscd	70.20	71.53	74.66	71.00	75.02	67.87	78.32
	Pre	69.47	68.17	77.38	68.59	74.37	72.23	76.89
	Rec	70.94	75.24	72.13	73.58	75.68	64.01	79.81
	OA	86.89	87.33	89.32	87.18	89.13	85.73	90.89
30%	mIoU	76.93	82.96	81.55	79.19	81.35	80.17	86.06
	Sek	31.42	43.61	40.72	33.93	40.32	36.45	50.57
	Score	45.07	55.42	52.97	47.50	52.63	49.57	61.22
	Fscd	72.10	79.11	77.82	73.05	77.41	74.20	82.47
	Pre	71.23	78.06	79.27	71.92	76.19	77.75	81.19
	Rec	72.99	80.18	76.41	74.22	78.67	70.96	83.80
	OA	87.73	91.01	90.59	88.51	90.16	88.77	92.77
40%	mIoU	79.38	85.80	82.95	81.43	84.48	81.08	87.53
	Sek	36.91	50.31	43.90	40.19	47.52	38.12	54.21
	Score	49.65	60.96	55.62	52.56	58.61	51.01	64.21
	Fscd	76.07	82.32	79.56	77.44	81.23	75.11	84.00
	Pre	78.13	80.94	81.16	77.52	80.56	78.13	83.13
	Rec	74.12	83.74	78.02	77.37	81.90	72.32	84.89
	OA	89.71	92.47	91.37	90.33	91.95	89.30	93.45

scheme that utilizes spatiotemporal constraints consistent with the SCD task to guide the learning of semantic changes and obtain semantic change detection results.

6) *SenseEarth2020:* This method uses a shared backbone and processes the structure of the respective segmentation task with two segmentation heads separately to obtain semantic change detection results.

This study was trained and tested on an NVIDIA RTX A6000 server based on the PyTorch framework. This model uses the AdamW optimizer to minimize the loss function with a weight decay coefficient of 0.0001 and a learning rate of 0.0003. To ensure convergence, the training batch size is set to 6, and the epoch is set to 100.

A. Data and Preprocessing

The SECOND dataset [45] is a high-resolution semantic change detection dataset with images from the cities of Hangzhou, Chengdu, and Shanghai, China. The image size is 512 × 512 pixels, with a resolution ranging from 0.3 to 5 m. It contains seven land cover types: unchanged category, water, ground, low vegetation, trees, buildings, and playgrounds. The dataset is divided into the training dataset (2076 pairs), test dataset (596 pairs), and validation dataset (296 pairs) according to a 70%–20%–10% split. The Landsat-SCD dataset [46] is a medium-resolution semantic change detection dataset, with images from Tumxuk City, Xinjiang, China, which is located on the edge of the Taklamakan Desert. The image size is 416 × 416 pixels, and the images are derived from Landsat data with a 30-m resolution from 1990 to 2020. It contains five land cover types: unchanged category, farmland, desert, building, and water. The dataset is divided into the training dataset (5924 pairs), test dataset (1698 pairs), and validation dataset (846 pairs) according to a 70%–20%–10% ratio. This study mainly focuses on semi-supervised semantic change detection; therefore, the training dataset is further divided into a labeled training dataset (with proportions of 10%, 20%, 30%, and 40%) and an unlabeled dataset.

B. Second Dataset

The results of the SCanNet method are less satisfactory due to the small number of 10% samples, as shown in Table I. The SenseEarth2020 method and the BT-HRSCD method have OA ranging from 79 to 84, SeK ranging from 1 to 15, and Fscd ranging from 37 to 54. The Bi-SRNet method, the HGINet method, and the CDSC method have OA ranging from 82 to 86, SeK ranging from 4 to 17, and Fscd ranging from 41 to 57. Among them, the accuracy of the CDSC method is slightly higher than that of the SCanNet, SenseEarth2020, BT-HRSCD, Bi-SRNet, and HGINet methods. All accuracy evaluation metrics of the MST-AK method are higher than those of the other compared methods, with OA ranging from 85 to 86, SeK ranging from 14 to 18, and Fscd ranging from 54 to 58, enabling it to achieve optimal semantic change detection results.

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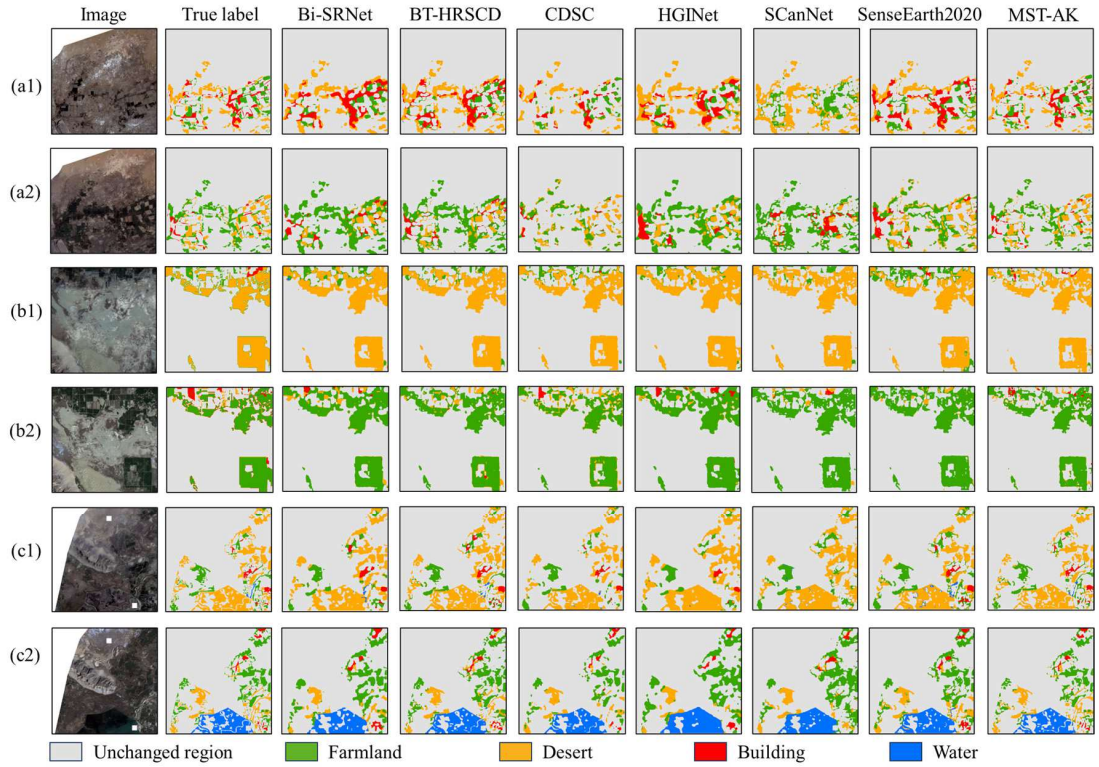


Fig. 9. Results of different methods for detecting changes in 10% of the samples in the Landsat-SCD dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

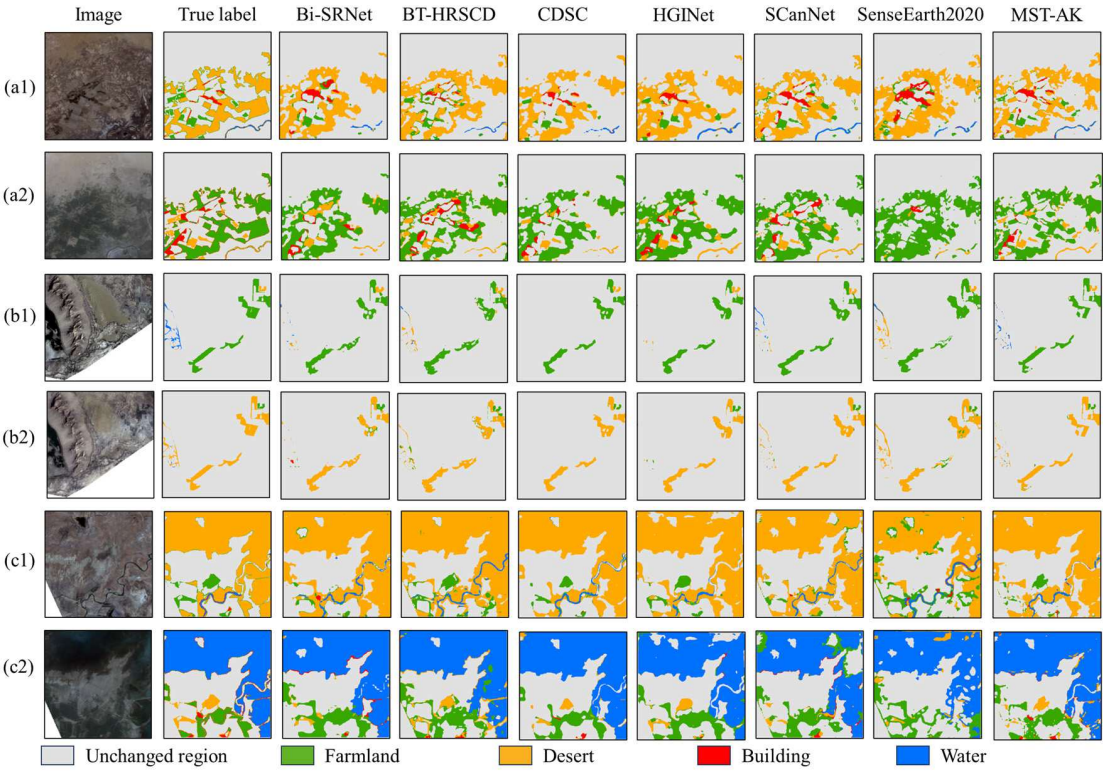


Fig. 10. Results of different methods for detecting changes in 20% of the samples in the Landsat-SCD dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

As shown in Figs. 5–8, at a 10%–40% sample ratio, the Bi-SRNet method produces false detections in identifying ground converted to trees and low vegetation converted to buildings. The BT-HRSCD method exhibits more significant

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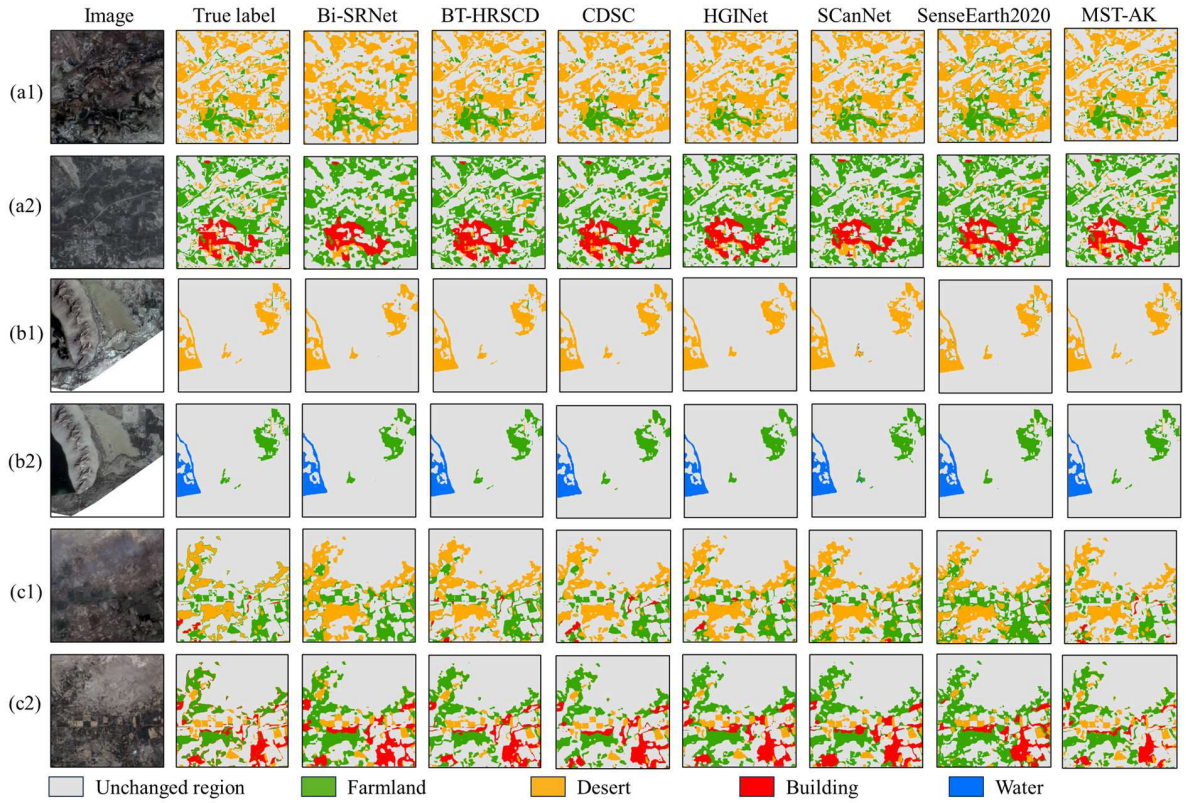


Fig. 11. Results of different methods for detecting changes in 30% of the samples in the Landsat-SCD dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

TABLE III

ACCURACY EVALUATION RESULTS OF ABLATION EXPERIMENTS FOR 10% SAMPLES ON SECOND DATASET

	Method	mIoU	Sek	Score	Fscd	Pre	Rec	OA
Method-1	Image feature	67.58	13.63	29.81	52.91	55.22	50.78	83.46
Method-2	Image feature + threshold	67.83	13.78	29.99	53.19	54.33	52.10	83.88
Method-3	Image feature + threshold + text feature	67.69	14.00	30.10	53.92	54.76	53.11	84.07
Method-4	Image feature + threshold + text feature + difference change feature fusion	67.89	14.05	30.20	54.20	51.36	57.37	85.09
Method-5	Image feature + threshold + text feature + difference change feature fusion + spatio-temporal association	67.78	14.32	30.36	54.30	50.22	59.10	85.40
Method-6	Image feature + threshold + text feature + difference change feature fusion + spatio-temporal association + image-text change difference Loss	68.02	14.06	30.25	54.01	49.83	58.96	85.44
MST-AK	Image feature + threshold + text feature + difference change feature fusion + spatio-temporal association + image-text change difference Loss + three-branch change difference Loss (MST-AK)	68.10	14.40	30.51	54.28	49.90	59.51	85.55

TABLE IV

COMPUTATIONAL EFFICIENCY OF DIFFERENT METHODS WITH 10% SAMPLES ON SECOND DATASET

	Bi-SRNet (epoch = 50)	BT-HRSCD (epoch = 50)	CDSC (epoch = 50)	HGINet (epoch = 80)	SCanNet (epoch = 50)	SenseEarth2020 (epoch = 100)	MST-AK (epoch = 100)
Parameter count	23313743	33328052	38595035	27702937	27897807	48314073	53389081
Running time/h	1.31	1.29	1.33	3.06	2.89	1.00	3.88

overdetection in detecting ground converted to buildings. The CDSC method and the HGINet method show substantial omissions in detecting low vegetation converted to buildings, while the HGINet method also has omissions in detecting ground converted to trees. The SCanNet method presents more pronounced overdetection and omissions. The SenseEarth2020 method incorrectly identifies ground converted to low vegetation when detecting ground converted to trees, and ground con-

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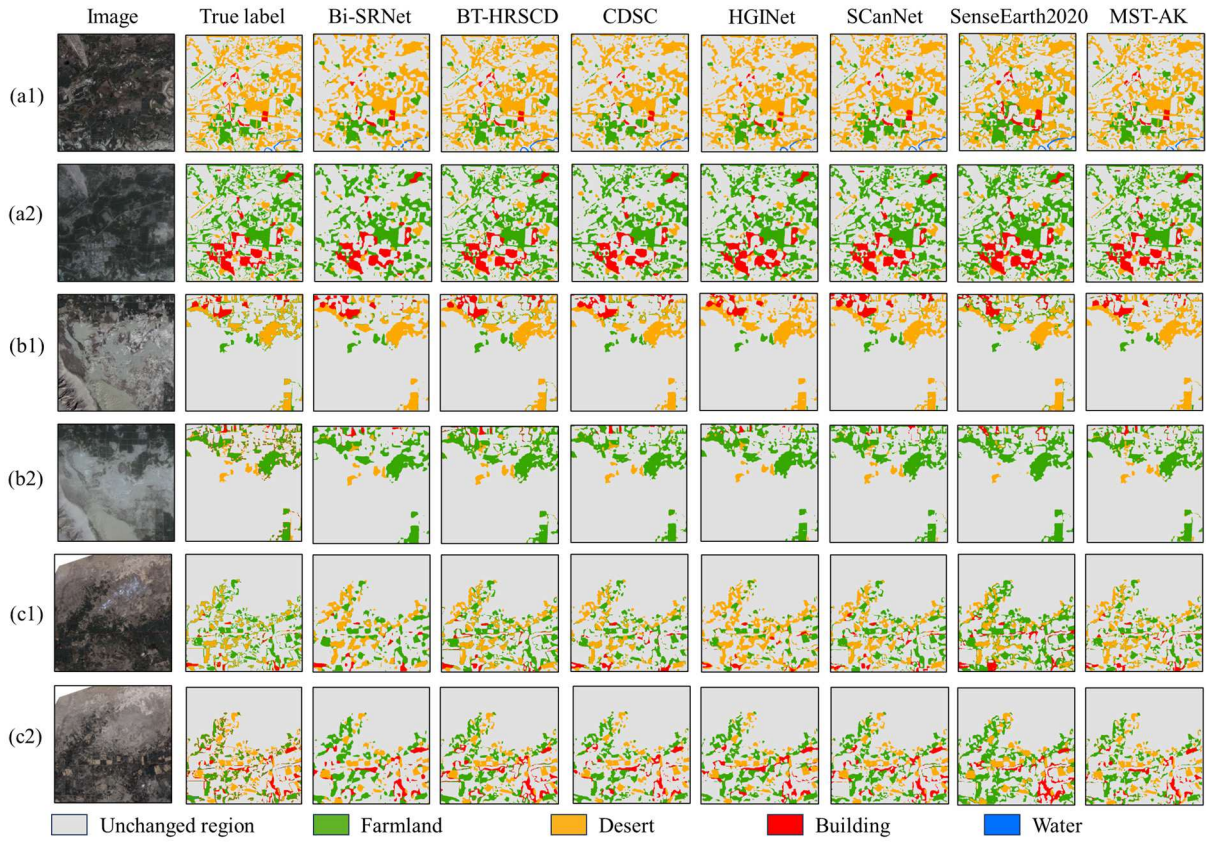


Fig. 12. Results of different methods for detecting changes in 40% of the samples in the Landsat-SCD dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

verted to trees when detecting ground converted to buildings. The MST-AK method is closest to the true semantic change detection results, although it still exhibits some omissions and misidentifications.

C. Landsat-SCD Dataset

As shown in Table II, under different sample ratios, the accuracy of the Bi-SRNet method, the HGINet method, the SCanNet method, and the SenseEarth2020 method are relatively close to each other, with the OA ranging from 82 to 91, SeK ranging from 20 to 47, and Fscd ranging from 61 to 81. The precision of the CDSC method and the BT-HRSCD method is higher than the above methods, with the OA ranging from 85 to 92, SeK ranging from 25 to 50, and Fscd ranging from 66 to 82. As the sample ratio increases, the precision of all the above methods significantly improves, especially for the BT-HRSCD and CDSC methods. Compared to the other methods, the MST-AK method achieves the highest accuracy at different sample ratios, with OA ranging from 87 to 93, SeK ranging from 30 to 54, and Fscd ranging from 71 to 84.

As shown in Figs. 9–12, in 10%–40% of the samples, changes from desert to farmland, buildings to farmland, and desert to water primarily occurred in the study area. Combined with the real semantic change labels, the Bi-SRNet method shows fuzzy boundaries for the semantic change detection results. The BT-HRSCD method exhibits overdetection for the desert-to-farmland conversion. The CDSC method shows omission detection for the building-to-desert and farmland conversion. The HGINet, SCanNet, and SenseEarth2020 methods all exhibit overdetection for desert-to-water conversion detection. The MST-AK method has fewer omissions and more accurate boundary identification than the other methods.

IV. Discussion

A. Ablation Experiments With MST-AK Method

To further illustrate the effectiveness of the MST-AK method, the SECOND dataset with a 10% sample ratio is used as an example for the corresponding ablation experiments. Method-1 obtains semantic change detection results using image features, combined with pretrained weights of the ResNet-34 network. Method-2 obtains semantic change detection results using image features, combined with pretrained weights of the ResNet-34 network and the change magnitude-based adaptive threshold selection method. Method-3 obtains semantic change detection results by using image features, combined with pretrained weights of the ResNet-34 network, the change magnitude-based adaptive threshold selection method, the CLIP-based descriptive cue textual knowledge module of the land cover type, and the image–text channel attention-based and image–text spatial attention-based feature extraction module. Method-4 obtains semantic change detection results using image features,

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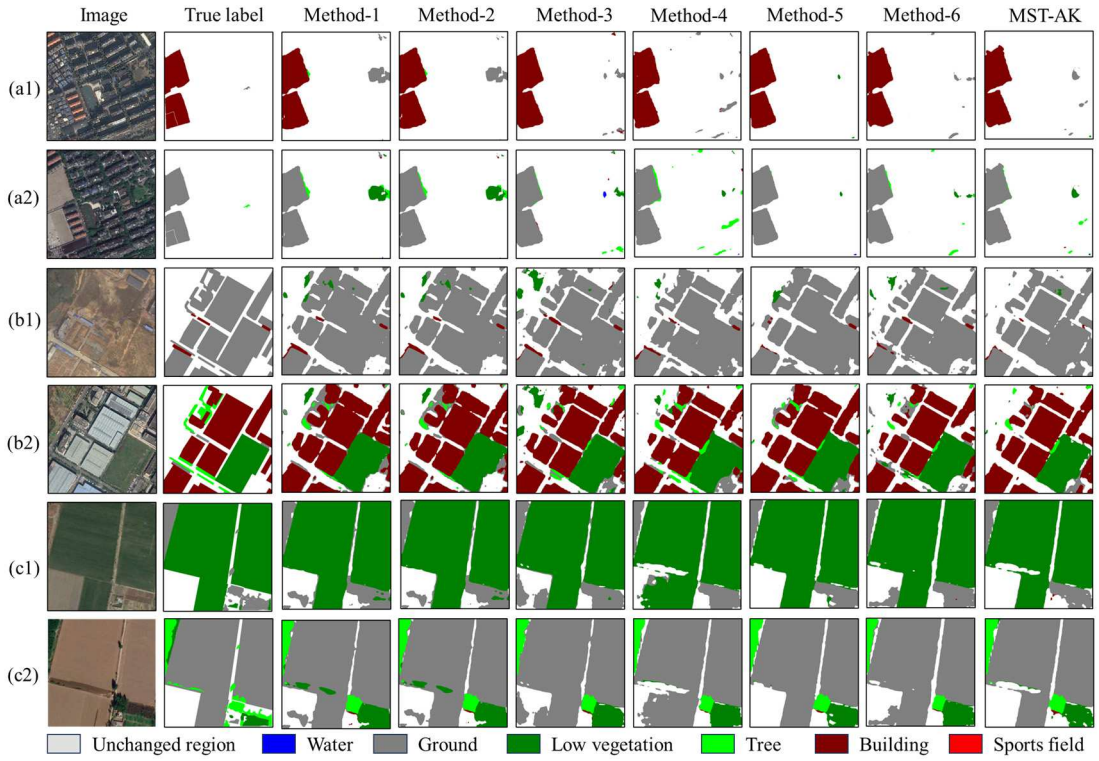


Fig. 13. Change detection results of ablation experiments for 10% samples on the SECOND dataset. (a1) and (a2), (b1) and (b2), and (c1) and (c2) are the semantic change detection results for three sets of different bi-temporal images, respectively.

TABLE V

ACCURACY EVALUATION OF LAND COVER SEMANTIC CHANGE DETECTION RESULTS IN BEIJING CITY AND YANGZHOU CITY, CHINA

Ratios	Size	Beijing City		Yangzhou City	
		256×256	512×512	256×256	512×512
10%	mIoU	71.10	71.44	84.68	84.08
	Sek	6.92	11.10	30.85	32.03
	Score	26.18	29.20	47.00	47.65
	Fscd	48.60	51.95	63.02	64.51
	Pre	51.52	51.71	60.35	64.21
	Rec	46.00	52.20	65.94	64.82
	OA	83.44	85.07	90.06	89.89
20%	mIoU	75.02	75.49	85.86	85.64
	Sek	23.95	26.74	38.43	38.37
	Score	39.27	41.37	52.66	52.55
	Fscd	62.95	66.13	69.82	70.06
	Pre	68.74	71.02	67.39	68.28
	Rec	58.07	61.87	72.43	71.93
	OA	86.91	87.89	92.01	91.94

combined with pretrained weights of the ResNet-34 network, the change magnitude-based adaptive threshold selection method, the CLIP-based descriptive cue textual knowledge module of the land cover type, the image–text channel attention-based and image–text spatial attention-based feature extraction module, and the difference change feature fusion representation module. Method-5 obtains semantic change detection results using image features, combined with pretrained weights of the ResNet-34 network, the change magnitude-based adaptive threshold selection method, the CLIP-based descriptive cue textual knowledge module of the land cover type, the image–text channel attention-based and image–text spatial attention-based feature extraction module, the difference change feature fusion representation module, and the three-branch interactive spatiotemporal association module. Method-6 obtains semantic change detection results using image features, combined with pretrained weights of the ResNet-34 network, the change magnitude-based adaptive threshold selection method, the CLIP-based descriptive cue textual knowledge module of the land cover type, the image–text channel attention-based and image–text spatial attention-based feature extraction module, the difference change feature fusion representation module, the three-branch interactive spatiotemporal association module, and the image–text change difference-based loss function computation module. Fig. 13 shows the results of semantic change detection

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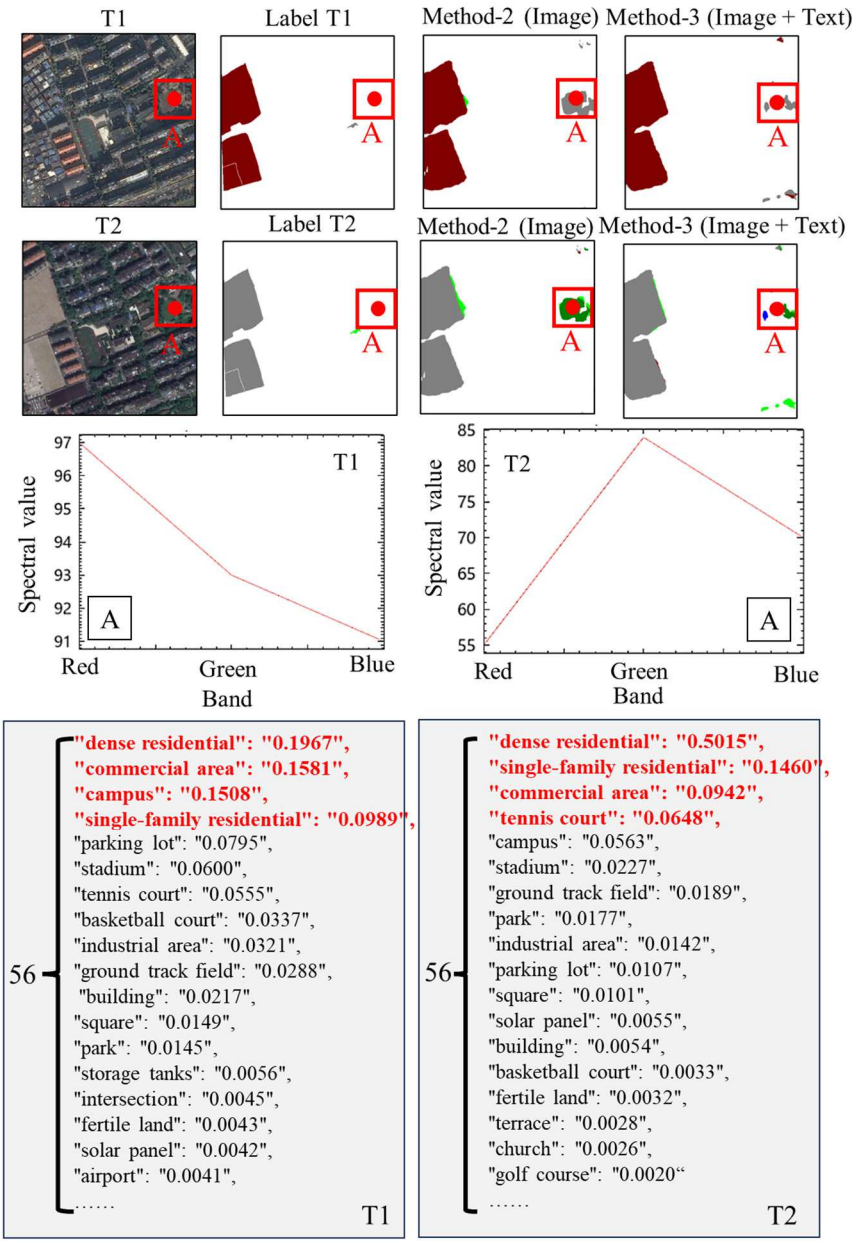


Fig. 14. When the spectral values of pixel point A are affected, CLIP-based descriptive cue textual knowledge of land cover types can supplement pixel point A.

for the MST-AK method ablation experiments at a 10% sample ratio. It can be clearly observed that Method-1 and Method-2 have partial misidentifications of ground to low vegetation in region (a1 and a2). Method-3, Method-4, Method-5, Method-6, and the MST-AK method exhibit relatively fewer such misidentifications. The ablation experiments of the MST-AK method in both region (b1 and b2) and region (c1 and c2) achieve decent semantic change detection results. As shown in Table III, the MST-AK method achieves the most satisfactory semantic change detection results overall, with an OA of 85.55, an SeK of 14.40, an Fscd of 54.28, and an mIoU of 68.10. The MST-AK method fully utilizes the multimodal features of image–text, deeply mines the three-branch spatiotemporal semantic correlations between bitemporal images, and combines them with the adaptive threshold selection method. This approach avoids the limitation of single-modal semantic change detection and improves the distinguishability of the detection results.

As shown in Table IV, the computational efficiencies of the different methods are compared on the SECOND dataset with a 10% sample ratio. It is clearly observed that, in terms of the number of model parameters, the MST-AK method has the largest number of parameters, followed by the SenseEarth2020 method, while the Bi-SRNet method has the smallest number of parameters. The epoch setting of the public code varies across methods, and in this study, we did not modify any parameters of the public code to avoid affecting the change detection results due to parameter modifications. In terms of runtime, the MST-AK method has the longest runtime, followed by the HGINet

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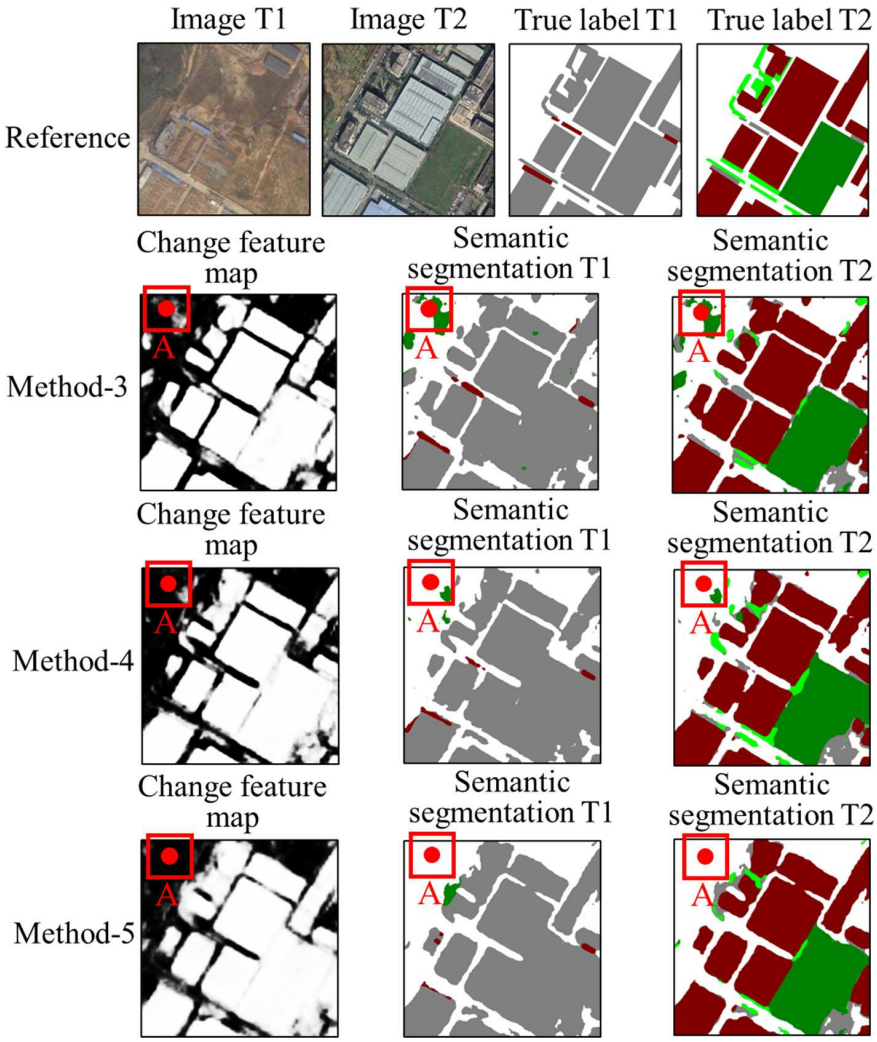


Fig. 15. Considering the difference change feature fusion and three-branch interactive spatiotemporal association can effectively prevent overdetection and underdetection in change detection results.

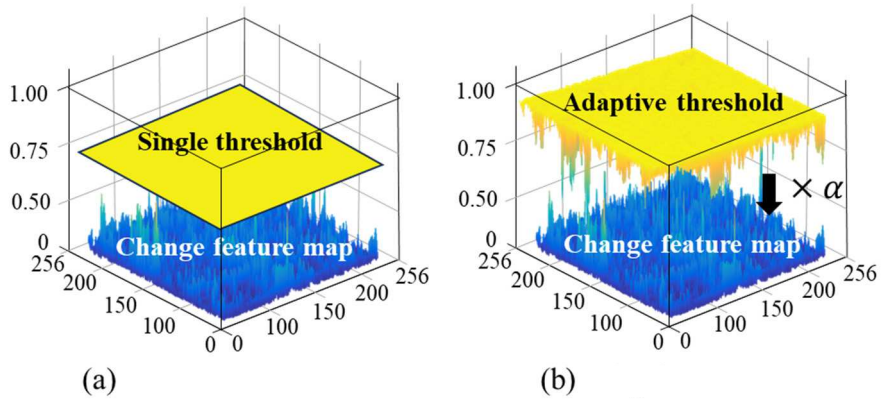


Fig. 16. Adaptive selection of threshold for change feature maps. Typically, change detection results are obtained using a single threshold. This study employs adaptive thresholds to achieve change detection results. (a) Single threshold. (b) Adaptive threshold.

method, while the SenseEarth2020 method has the shortest runtime.

B. Robustness of MST-AK Method

The robustness of the MST-AK method is further demonstrated in terms of image–text multimodal semantic features, differential change feature fusion, three-branch interactive spatiotemporal association, and adaptive threshold selection.

The MST-AK method captures deep semantic features from an image–text multimodal perspective, effectively avoiding the problem of underutilization of single-modal semantic feature

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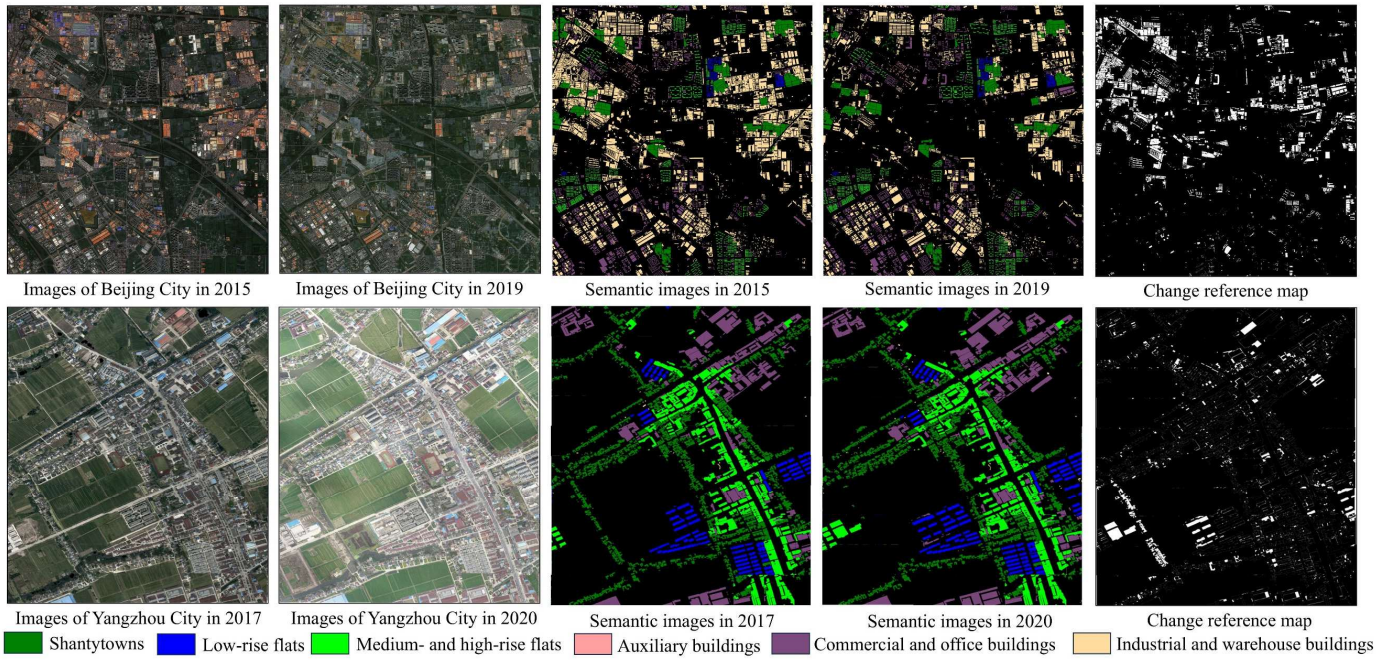


Fig. 17. Semantic change detection of land cover in selected areas of Beijing City and Yangzhou City, China.

information, and improving the accuracy of semantic change detection results. As shown in Fig. 14, the land cover type of pixel point A in the image is classified as ground. Pixel point A is affected by external factors, and its RGB band spectral values and spectral curve shapes exhibit significant differences. If only the image spectral feature information is considered, excessive detections may occur, which in turn affects the accuracy of the semantic change detection results. The CLIP-based descriptive cue textual knowledge of land cover types describes the image as dominated by types such as dense residential, commercial area, and single-family residential, among others. This helps to focus more on these land cover types during semantic change detection, thereby avoiding overdetection and underdetection. Method-2 considers only image feature information, which leads to the misidentification of ground to low vegetation at pixel point A. Method-3, however, incorporates both image–text multimodal depth features, which helps to avoid overdetection and underdetection. Therefore, the MST-AK method is more stable and reliable compared to single-modal semantic change detection methods, as it fully considers both image and text features to construct multimodal deep semantic features.

As shown in Fig. 15, no land cover type change occurs at pixel point A in the image. The change magnitude value of pixel point A in the change feature map of Method-3 is high, and when the threshold is applied to obtain the semantic change detection result, overdetection easily occurs, resulting in a false identification of low vegetation at pixel point A. In contrast, the change magnitude value of pixel point A in the change feature map of Method-4 is low, and applying the threshold to obtain the semantic change detection result does not produce any false type identification. By using the difference change feature fusion representation module, multiscale spatiotemporal semantic change features are fused from different perspectives (i.e., feature splicing, absolute difference, and cosine similarity) to realize temporal multilevel feature complementarity and enhance the comprehensiveness of semantic change feature extraction. Although there is no land cover type change around pixel point A, the change feature map for Method-4 shows a high change magnitude value in the surrounding area, which still leads to a small number of misidentified regions. In contrast, the change feature map of Method-5 does not have high change magnitude values around pixel point A, and no false types are detected. Therefore, the MST-AK method cannot only extract comprehensive semantic change features but also avoid the limitations of single-branch models that tend to ignore spatiotemporal associations, demonstrating the model's spatiotemporal interaction capability.

Typically, a single threshold is applied to the change feature map to obtain change detection results. However, this method may lead to thresholds that are not adaptive. Method-1 and Method-2 perform semantic change detection using a single threshold and adaptive thresholds, respectively. The accuracy evaluation indexes of Method-1 are generally lower than those of Method-2. As shown in Fig. 16, the change magnitude-based adaptive threshold selection method adaptively selects thresholds based on the change magnitude values of each pixel in the change feature map—smaller thresholds for changed regions and larger thresholds for unchanged regions. Consequently, the MST-AK method improves the differentiation between changed and unchanged regions, thereby enhancing the accuracy of semantic change detection results.

C. Demonstration of Change Detection Applications

The MST-AK method is applied to semantic change detection in selected areas of Beijing City and Yangzhou City,

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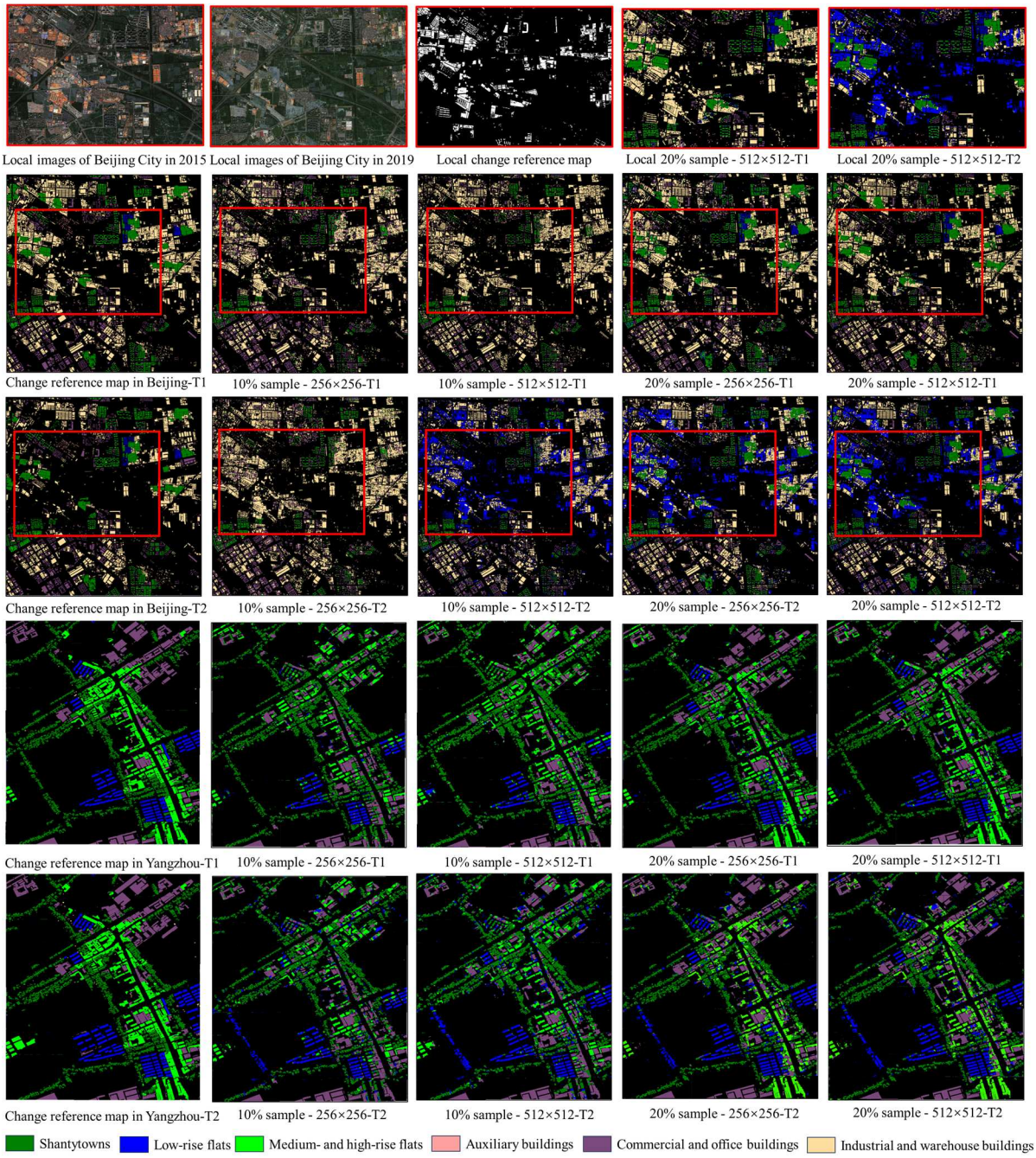


Fig. 18. Results of the MST-AK method for semantic change detection of land cover in Beijing City and Yangzhou City, China.

China. As shown in Fig. 17, there are significant differences in the land cover semantic change detection data of Beijing City and Yangzhou City, such as variations in data sources, building morphology, and spatial distribution. The land cover semantic change detection data for Beijing City consist of Gaofen-2 images from 2015 to 2019, with an image size of $9785 \times 10\,061$ pixels and a spatial resolution of 1 m. It includes six building types: shantytowns, low-rise residences, medium- and high-rise residences, commercial and office buildings, industrial and warehousing buildings, and auxiliary buildings, with a total of 19 000 buildings delineated. The land cover semantic change detection data for Yangzhou City consist of three pairs of aerial images from 2017 to 2020, with dimensions of $33\,774 \times 23\,946$, $36\,927 \times 29\,164$, and $26\,238 \times 24\,841$ pixels, with a spatial resolution of 0.1 m. The data for Yangzhou City include the same land cover types as Beijing City, with a total of 20 544 buildings classified [47]. In this study, the land cover semantic change detection data for both cities are cropped to 256×256 pixels and 512×512 pixels, with 10% and 20% of the samples are selected for semantic change detection.

As shown in Table V, different image cropping sizes affect the accuracy of the semantic change detection results of the MST-AK method. It can be clearly observed that the MST-AK method produces good semantic change detection results across different sample ratios and image cropping scales.

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Specifically, in Beijing City, the semantic change detection results range from 83 to 87 for OA, 6–26 for SeK, 71–75 for mIoU, 48–66 for Fscd, 26–41 for Score, 51–71 for Pre, and 46–61 for Rec. In Yangzhou City, the results of semantic change detection range from 89 to 92 for OA, 30–38 for SeK, 84–85 for mIoU, 63–70 for Fscd, 47–52 for Score, 60–68 for Pre, and 64–72 for Rec. With an increase in the sample ratio, the semantic change detection results of the MST-AK method show a significant improvement.

As shown in Fig. 18, at sample ratios of 10% and 20%, different image cropping sizes have some influence on the semantic change detection results. As the sample ratio increases, the semantic change detection results of the MST-AK method become closer to the reference semantic change detection results. Notably, in the red region, there is an obvious shift from industrial and warehousing buildings to low-rise residences in Beijing City. The MST-AK method, by deeply mining spatiotemporal semantic change features from the image–text multimodal perspective and utilizing the three-branch interaction spatiotemporal association along with the adaptive threshold selection method, effectively avoids omissions and misidentifications. It therefore better captures the changes in land cover types and achieves cross-modal semantic change detection.

D. Limitations of the MST-AK Method

The MST-AK method achieves decent results in semantic change detection, but there is still room for improvement. First, its computational efficiency is lower than that of comparative methods. To balance efficiency and accuracy, future research may consider optimizing algorithms by employing parallel computing or other acceleration techniques or by reducing unnecessary computational steps to enhance operational speed. Second, the MST-AK method performs semantic change detection at the image pixel level, and the "salt and pepper" phenomenon still exists. The accuracy of the semantic change detection results should be further improved at the image object level. Furthermore, the MST-AK method detects semantic changes in bitemporal remote sensing images, and the rich time series of remote sensing images is a distinguishing feature compared to other types of images. The semantic change features of remote sensing images based on time series should be explored further to enhance the accuracy of the semantic change detection results. Finally, the MST-AK method is implemented locally, but future efforts should focus on encapsulating it on a cloud platform to enable the study of long-time series semantic change detection.

V. Conclusion

In this study, the MST-AK method is proposed, which fully exploits the three-branch interaction spatiotemporal semantic change features between different temporal images from an image–text multimodal perspective, and effectively avoids the problem of the limited spatiotemporal semantic extraction capability of unimodal methods. Furthermore, the MST-AK method improves the distinguishability between changed and unchanged regions by using the change magnitude-based adaptive threshold selection method, resulting in better semantic change detection results.

Comparative analysis of the MST-AK method with state-of-the-art methods, along with corresponding ablation experiments, is conducted to qualitatively and quantitatively assess the usability of the MST-AK method using the publicly available SECOND dataset and Landsat-SCD dataset. In both the SECOND and Landsat-SCD datasets, the OA of the MST-AK method is improved by about 1%–5%, the SeK is improved by about 3%–13%, and the Fscd is improved by about 3%–11% with different sample ratios compared to the state-of-the-art methods. Based on this, the effectiveness of the MST-AK method is further demonstrated using data from parts of Beijing City and Yangzhou City, China, to showcase the application of semantic change detection.

A potential future research direction is to integrate the MST-AK method into a cloud platform for long-term time series semantic change detection of land cover, in order to meet practical application needs in fields such as land cover dynamic monitoring, urban land spatial planning, and disaster hazard monitoring.

Acknowledgment

The authors would like to thank the editors and the anonymous reviewers for their constructive comments and suggestions, which greatly helped to improve the quality of the article.

References

[1] M. Zhao et al., "Spatially and semantically enhanced Siamese network for semantic change detection in high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2563–2573, 2022.

[2] B. Fang, G. Chen, G. Ouyang, J. Chen, R. Kou, and L. Wang, "Content-invariant dual learning for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603317.

[3] S. Du, S. Du, B. Liu, and X. Zhang, "Incorporating DeepLabv3+ and object-based image analysis for semantic segmentation of very high resolution remote sensing images," *Int. J. Digit. Earth*, vol. 14, no. 3, pp. 357–378, Mar. 2021.

[4] S. Du et al., "Mapping urban functional zones with remote sensing and geospatial big data: A systematic review," *GISci. Remote Sens.*, vol. 61, no. 1, Dec. 2024, Art. no. 2404900.

[5] L. Zhu, H. Xing, and D. Hou, "Analysis of carbon emissions from land cover change during 2000 to 2020 in Shandong Province, China," *Sci. Rep.*, vol. 12, no. 1, pp. 1–12, May 2022.

[6] L. Zhu, W. Sun, D. Fan, H. Xing, and H. Ban, "Exploring the level of agricultural development using greenhouse mapping: A case study of Shandong Province, China," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 5793–5809, 2025.

[7] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[8] Q. Xia et al., "SSTNet: A network based on modeling of semantic-spatiotemporal information for semantic change detection of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 18365–18378, 2024.

[9] X. Zuo et al., "Multitask Siamese network guided by enhanced change information for semantic change detection in bitemporal remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 61–77, 2025.

[10] L. Zhu, H. Xing, L. Zhao, H. Qu, and W. Sun, "A change type determination method based on knowledge of spectral changes in land cover types," *Earth Sci. Informat.*, vol. 16, no. 2, pp. 1265–1279, Jun. 2023.

[11] L. Zhu, W. Sun, Q. Zhang, C. Wang, and Z. Guo, "Fine-grained agricultural and pastoral information extraction using Sentinel-1 and Sentinel-2 intra-year time series in Jingyang district, Deyang City," *Adv. Space Res.*, vol. 72, no. 9, pp. 4031–4047, Nov. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:44:37 UTC from IEEE Xplore. Restrictions apply.

[12] Y. Zhang et al., "Mapping causal agents of disturbance in boreal and Arctic ecosystems of North America using time series of Landsat data," *Remote Sens. Environ.*, vol. 272, Apr. 2022, Art. no. 112935.

[13] Y. He, H. Zhang, X. Ning, R. Zhang, D. Chang, and M. Hao, "Spatial–temporal semantic perception network for remote sensing image semantic change detection," *Remote Sens.*, vol. 15, no. 16, pp. 4095–4118, Aug. 2023.

[14] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[15] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[16] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, and P. He, "SCDNET: A novel convolutional network for semantic change detection in high resolution optical remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102465.

[17] H. Xia, Y. Tian, L. Zhang, and S. Li, "A deep Siamese postclassification fusion network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622716.

[18] R. Caye Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Multitask learning for large-scale semantic change detection," *Comput. Vis. Image Understand.*, vol. 187, Oct. 2019, Art. no. 102783.

[19] L. Ding, H. Guo, S. Liu, L. Mou, J. Zhang, and L. Bruzzone, "Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620014.

[20] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.

[21] P. Chen, B. Zhang, D. Hong, Z. Chen, X. Yang, and B. Li, "FCCDN: Feature constraint network for VHR image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 101–119, May 2022.

[22] Y. Niu, H. Guo, J. Lu, L. Ding, and D. Yu, "SMNet: Symmetric multi-task network for semantic change detection in remote sensing images based on CNN and transformer," *Remote Sens.*, vol. 15, no. 4, p. 949, 2023.

[23] J. Long, M. Li, X. Wang, and A. Stein, "Semantic change detection using a hierarchical semantic graph interaction network from high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 318–335, May 2024.

[24] L. Tan and X. Zuo, "CGMNet: Semantic change detection via a change-aware guided multi-task network," *Remote Sens.*, vol. 16, no. 13, pp. 1–19, 2024.

[25] L. Mei et al., "SCD-SAM: Adapting segment anything model for semantic change detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5626713.

[26] S. Du, S. Du, B. Liu, and X. Zhang, "Mapping large-scale and fine-grained urban functional zones from VHR images using a multi-scale semantic segmentation network and object based approach," *Remote Sens. Environ.*, vol. 261, Aug. 2021, Art. no. 112480.

[27] Z. Wang, Z. Pan, H. Long, and Y. Hu, "Enhanced multitask semantic change detection via semi-supervised learning in LULC segmentation subtask," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] K. Heidler, I. Nitze, G. Grosse, and X. X. Zhu, "PixelDINO: Semi-supervised semantic segmentation for detecting permafrost disturbances in the Arctic," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4302512.

[29] P. Lv, P. Cheng, C. Ma, and Y. Zhong, "A semi-supervised semantic and spatial change detail retention network for semantic change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4416516.

[30] X. Zhang et al., "Remote sensing image semantic change detection boosted by semi-supervised contrastive learning of semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624113.

[31] G. Chen, B. Shi, Y. Zhang, Z. He, and P. Zhang, "CGSNet: Cross-consistency guiding semi-supervised semantic segmentation network for remote sensing of Plateau lake," *J. Netw. Comput. Appl.*, vol. 230, Oct. 2024, Art. no. 103974.

[32] A. Dai et al., "A difference enhancement and class-aware rebalancing semi-supervised network for cropland semantic change detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 137, Mar. 2025, Art. no. 104415.

[33] Z. Li, B. Chen, S. Wu, M. Su, J. M. Chen, and B. Xu, "Deep learning for urban land use category classification: A review and experimental assessment," *Remote Sens. Environ.*, vol. 311, Sep. 2024, Art. no. 114290.

[34] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.

[35] S. Wang, Y. Duan, H. Ding, Y.-P. Tan, K.-H. Yap, and J. Yuan, "Learning transferable human-object interaction detector with natural language supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 929–938.

[36] V. Zermatten, J. Castillo-Navarro, D. Marcos, and D. Tuia, "Learning transferable land cover semantics for open vocabulary interactions with remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 621–636, Feb. 2025.

[37] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, San Francisco, CA, USA, 2021, pp. 1–48.

[38] H. Xing et al., "An adaptive change threshold selection method based on land cover posterior probability and spatial neighborhood information," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11608–11621, 2021.

[39] L. Zhu, W. Sun, D. Fan, H. Xing, and X. Liu, "Unsupervised spatial self-similarity difference-based change detection method for multi-source heterogeneous images," *Pattern Recognit.*, vol. 149, pp. 110237–110259, May 2024.

[40] L. Zhu, W. Sun, H. Xing, D. Fan, and C. Wang, "A phenological knowledge transfer-based fine grained land cover change sample collection method: A case study of coastal wetlands," *Int. J. Digit. Earth*, vol. 17, no. 1, pp. 2310090–2310109, Dec. 2024.

[41] L. Zhu, Z. Guo, H. Xing, and W. Sun, "A coupled temporal–spectral–spatial multidimensional information change detection framework method: A case of the 1990–2020 Tianjin, China," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5741–5758, 2023.

[42] S. Fang, W. Li, S. Yang, Z. Li, J. Zhao, and X. Wang, "BT-HRSCD: High-resolution feature is what you need for a semantic change detection network with a triple-decoding branch," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4416714.

[43] Q. Wang, W. Jing, K. Chi, and Y. Yuan, "Cross-difference semantic consistency network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4406312.

[44] L. Ding, J. Zhang, H. Guo, K. Zhang, B. Liu, and L. Bruzzone, "Joint spatio-temporal modeling for semantic change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610814.

[45] K. Yang et al., "Asymmetric Siamese networks for semantic change detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609818.

[46] P. Yuan, Q. Zhao, X. Zhao, X. Wang, X. Long, and Y. Zheng, "A transformer-based Siamese network and an open optical dataset for semantic change detection of remote sensing images," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 1506–1525, Dec. 2022.

[47] Q. Shen, S. Tao, R. Yang, X. Zhang, and M. Wang, "BSCDNet: A building change detection network with category differentiation using a graph attention mechanism and multitask learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 15761–15776, 2024.

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